

Heaven's Light is Our Guide



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Rajshahi University of Engineering & Technology, Bangladesh

**Subspace Detection for Hyperspectral Image Classification Using
Folded-PCA and Cross Cumulative Residual Entropy**

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CERTIFICATE

*This is to certify that this thesis report entitled **Subspace Detection for Hyperspectral Image Classification Using Folded-PCA and Cross Cumulative Residual Entropy** submitted by **Md. Iftekharul Islam, Roll: 133096**, in fulfillment of the requirement for the award of the degree **Bachelor of Science in Computer Science & Engineering of Rajshahi University of Engineering & Technology, Bangladesh** is a record of the candidate's own work carried out by him under my supervision. This thesis has not been submitted for the award of any other degree.*

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Abstract

Hyperspectral image (HSI) analysis has become an interesting research area because of its various applications. Finding a proper subspace is very important to effectively classify hyperspectral image since the bands in a hyperspectral image are highly correlated. Feature reduction is often performed using feature extraction and feature selection. Principal Component Analysis (PCA) is a widely used feature extraction method. But PCA suffers from high computation cost, large memory requirement and low efficacy in dealing with hyperspectral images. Folded-PCA (FPCA) overcomes these problems that considers the local structures of the data. In this thesis, a combined approach is proposed where FPCA is applied to generate new features from the original spectra bands. Then feature selection is performed on the extracted features using Cross Cumulative Residual Entropy (CCRE) to select the relevant features. The experimental result shows that the classification accuracy of the proposed method (FPCA-CCRE) is 97.9739% outperforming FPCA with classification accuracy of 97.7146% and the combination of FPCA and normalized Mutual Information (FPCA-nMI) with classification accuracy of 97.9282%.

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Chapter 1

Introduction

1.1 Overview

Hyperspectral sensors are a systems technology which collect images of a scene in tens to hundreds of narrow spectral bands simultaneously. The pixels of the captured images form a three-dimensional hypercube providing more information on the scene [1]. For example, hyperspectral data cubes generated from the NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) contains 224 bands with a fine spectral resolution of $0.01\mu\text{m}$ [2]. The high data volume enriched with information also presents various challenges. The data captured are highly correlated and contains a significant amount of redundant data. All the image bands may not be equally important for a specific application. Also, as the feature space dimension increases, the classification accuracy starts to decrease if the size of the training data does not grow correspondingly. This effect is known as the Hughes phenomenon [3]. Therefore developing effective techniques to find relevant features from input dataset is a major interest in current literature. Principal component analysis (PCA) [4] is a popular feature extraction technique although it has some limitations. It considers only the global variance of the dataset and faces memory management issues for dataset with large spatial size [5]. Folded-PCA (FPCA) is a method that overcomes these problems. There are few techniques for finding relevant features in current literature such as Mutual Information (MI) and Cross Cumulative Residual Entropy (CCRE) [6]. In this thesis work, a feature mining technique that combines FPCA and CCRE is proposed for effective subspace detection. Major focus is given on how well the proposed collaboration of feature extraction and feature selection works while utilizing support vector machine (SVM) [7] to achieve better classification accuracy.

1.2 Remote Sensing

Remote sensing is a technology of collecting and analyzing data to get information about an object without the instrument used to collect the data being in direct contact with the object. Remote sensing is used in numerous fields, including geography, land surveying and most Earth Science disciplines (for example, hydrology, ecology, oceanography, glaciology, geology); it also has military, intelligence, commercial, economic, planning, and humanitarian applications. In current usage, the term "remote sensing" generally refers to the use of satellite- or aircraft-based sensor technologies to detect and classify objects on Earth, including on the surface and in the atmosphere and oceans, based on propagated signals (e.g. electromagnetic radiation).

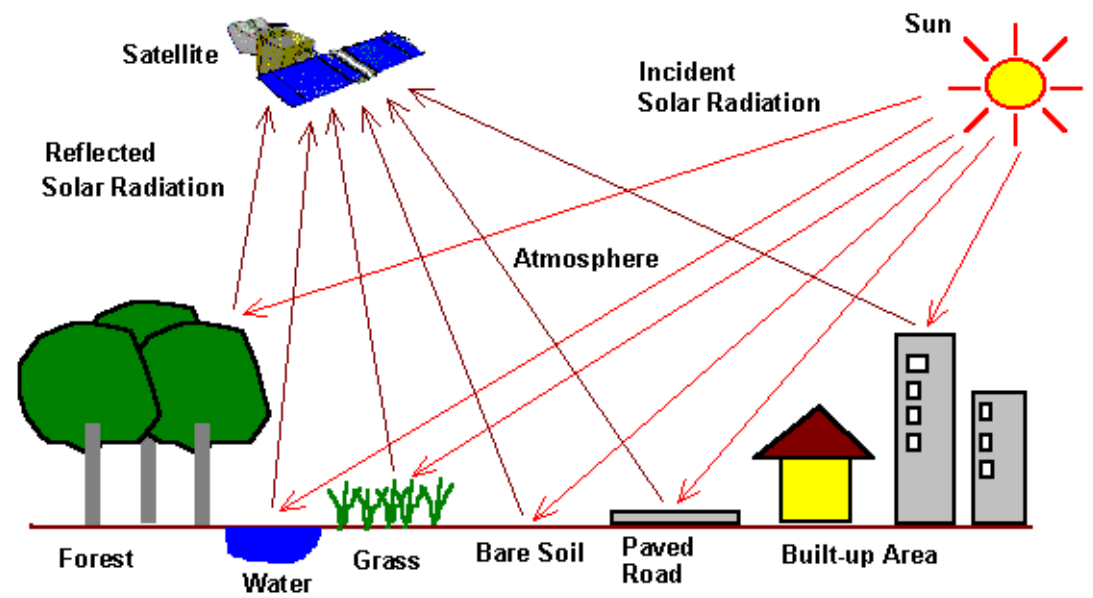


Figure 1.1: Remote sensing [8]

Remote sensors can be either passive or active. Passive sensors respond to external stimuli. They record natural energy that is reflected or emitted from the Earth's surface. The most common source of radiation detected by passive sensors is reflected sunlight. In contrast, active sensors use internal stimuli to collect data about Earth. For example, a laser-beam remote sensing system projects a laser onto the surface of Earth and measures the time that it takes for the laser to reflect back to its sensor.

1.3 Challenges of Hyperspectral Data Processing

Hyperspectral imaging is a fast growing field that takes advantages of optical spectroscopy as an analytical tool combined with two-dimensional object visualization obtained by optical imaging. Hyperspectral data provide a very fine spectral characterization of the sensed materials, which facilitates their detection and categorization. Although advances in hardware to acquire hyperspectral imagery have made such data easily accessible to a wide variety of application domains, they have also created unique challenges in the analysis of such data. Some of these challenges are given below.

1.3.1 Data Size

The abundant information come with a huge amount of data and it is one of the major issues in hyperspectral data analysis. Traditional information processing techniques cannot be used to process such data effectively.

1.3.2 High Data Redundancy

The images contained in a hyperspectral data cube often have significant overlap of information among themselves. Consequently, many redundant bands are included in the typical collection of hundreds of spectral bands. Fig. 1.2 shows correlations among bands of AVIRIS Indian Pines data set where the white color indicates highly correlated bands with redundant information.

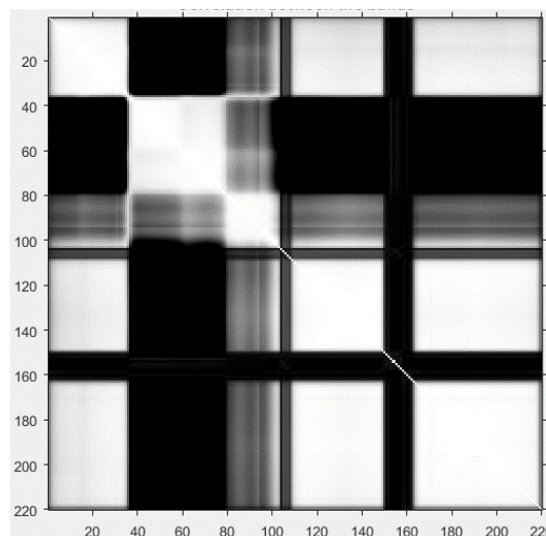


Figure 1.2: Correlation among bands of Indian Pines data set

1.3.3 Hughes Phenomenon

In 1968, a theoretical basis for the relationship between the number of features and the number of training samples with respect to generalization was presented by Hughes [3]. As the number of spectral bands increases, the classification accuracy starts to decrease after a certain point if the size of the training data does not grow correspondingly. This effect is known as the Hughes phenomenon or the curse of dimensionality.

There are two major observations to be made here. First, even with an infinite set of training samples, there is a saturation effect after a certain number of features. Second, the accuracy decreases after reaching a maximum. An intuitive explanation of this curse of dimensionality might be as follows. As the dimensionality increases, so does the complexity of the statistics that govern the classes. The training portion of a classification method involves the estimation of these statistics. Since only a finite number of training samples is available, the more complex the statistics are, the less accurate the estimation will be. This combination of a less accurate estimate of the statistics with a saturating separability of the classes, is the reason why the number of features must be limited. Adding the fact that, neighboring bands in hyperspectral data are generally strongly correlated, at first sight it may seem that multispectral data is sufficient for most applications. However, the required spectral bands for classification purposes should be specifically adapted to the problem at hand, and rarely correspond to the band settings of any multispectral sensor.

1.4 Motivation

Hyperspectral image processing has become an emerging research field because of its various applications. Remotely sensed ground areas are classified into different groups of object by classification of the hyperspectral image. The classification of the high-dimensional data is a time consuming task and it requires a significant computational capability which may not be available in many cases. The challenges of hyperspectral image processing motivate us to find the optimal subspace of the hyperspectral data to obtain efficient and reliable classification. Therefore, feature extraction and selecting only relevant features before classification is an important task to achieve high classification accuracy.

1.5 Objective

The main objective of this research work is to present a subspace detection method which combines feature extraction using folded principal component analysis (FPCA) and relevant feature selection by cross cumulative residual entropy (CCRE) to achieve a better classification accuracy.

1.6 Thesis Outline

This thesis is organized into 5 chapters. A brief description of the chapters is given as follows: In chapter 1, a brief overview of the whole research work and topics related to this research is discussed.

In chapter 2, The basic concepts of hyperspectral remote sensing, sensors, applications of hyperspectral images are discussed.

Chapter 3 focuses on the current literature of this research field. Recent and relevant feature mining techniques are discussed in this chapter.

In chapter 4, features extraction based on FPCA and relevant features selection using CCRE, their implementation with experimental results on real hyperspectral image are discussed in this chapter. Detailed explanations, algorithm, flow chart and experimental results of our proposed subspace detection method is also discussed in this chapter. Chapter 4 is the heart of this thesis report book.

Finally, chapter 5 contains a brief summary and concluding remarks. It also gives recommendations for further research.

Chapter 2

Fundamentals of Hyperspectral Imaging

2.1 Introduction

The advancement of remote sensing technology is always focused on achieving higher spectral and spatial resolutions. Hyperspectral sensor systems represent a revolutionary breakthrough, increasing the spectral resolution of sensors to as high as a few nanometres. This means that they are capable of generating nearly continuous spectral profiles for land surface materials. A hyperspectral sensor system is an imaging spectrometer which combines the spatial imaging capability of an imaging system with the spectral analytical tools of a spectrometer. Such a sensor system may have up to several hundred narrow spectral bands with spectral resolution on the order of 10 nm or narrower [1]. Imaging spectrometers produce a complete spectrum for every pixel in the image. The resulting dataset allows identification of materials rather than mere discrimination as with broadband sensor systems. This chapter gives a brief discussion on some of the basic concepts related to remote sensing to facilitate a better understanding of hyperspectral imagery.

2.2 Radiation and Reflectance

Looking at the bright sun we can observe the electromagnetic (EM) radiation emitted. This form of energy is transferred through space in the form of an oscillating sine wave, composed of different wavelengths, corresponding to the EM spectrum. For example, visible light has wavelengths between 0.4 and 0.7 microns, while radio waves have wavelengths greater than about 30 cm (Figure 2.1).

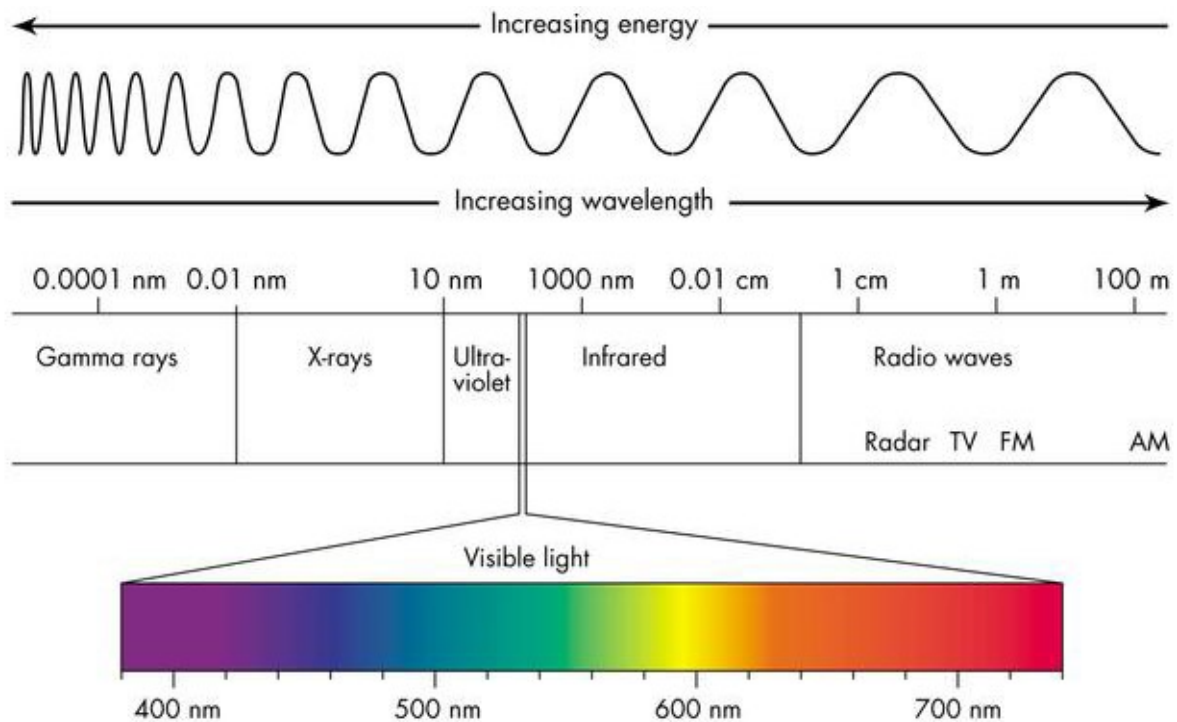


Figure 2.1: The electromagnetic spectrum [9]

Bodies that are not black only absorb a proportion of the incident radiation, defined by the absorptance. Reflectance is the percentage of the light hitting a material that is then reflected by that material. The nature of reflection depends on sizes of surface roughness or smoothness in relation to the wavelength of the radiation considered. A reflectance spectrum shows the reflectance of a material measured across a range of wavelengths (Figure 2.2).

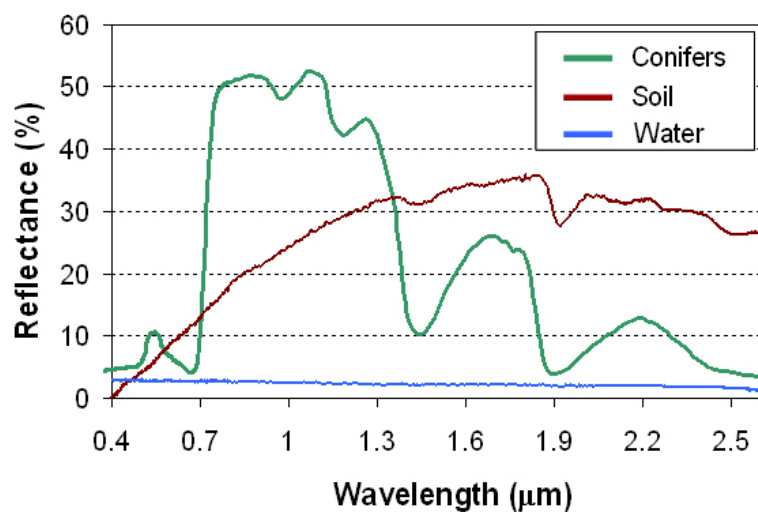


Figure 2.2: Reflectance spectra for 3 different types of features on the surface of the Earth [10]

Field and laboratory spectrometers usually measure reflectance at many narrow, closely spaced wavelength bands, so that the resulting spectra appear to be continuous curves (Fig. 2.2). When a spectrometer is used in an imaging sensor, the resulting images record a reflectance spectrum for each pixel in the image (Fig. 2.3).

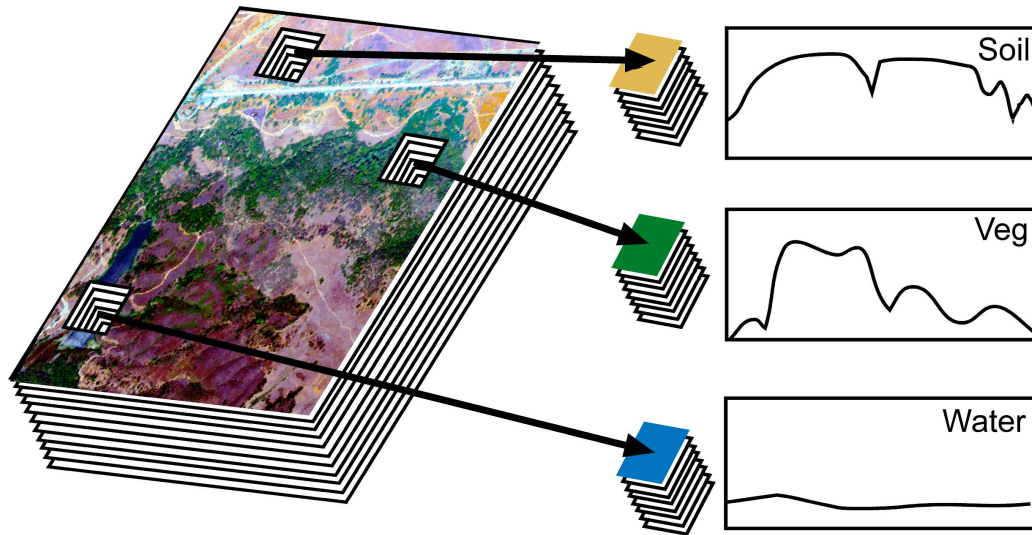


Figure 2.3: Hyperspectral data cube [2]

2.3 Hyperspectral versus Multispectral Imaging

Remotely sensed data acquisition from the surface of the Earth is a timely and effective way for gathering information about the natural and built environments for planning and management of human resources. Since the 1960s, multispectral imagery has been used from satellite systems and airborne for this purpose. Multispectral imagers, such as Landsat and SPOT, measure radiation reflected from a surface at a few wide, separated wavelength bands. This limits their functionality for Earth observation purposes. On the other hand, hyperspectral imagers measure reflected radiation at a series of narrow and contiguous wavelength bands.

As shown in Fig.2.3, each pixel in a hyperspectral image is represented as a high-dimensional vector containing values corresponding to reflectance spectrum so that the size of the vector is equal to the number of spectral bands. Therefore, the hyperspectral image can be considered as a three dimensional data cube that stacks multiple gray scale images together corresponding to different spectral channels from the same scene. A spectrum for one pixel in a hyperspectral image looks very much like a spectrum that would be measured in a spectroscopy laboratory.

This type of detailed pixel spectrum can provide much more information about the surface than a multispectral pixel spectrum. Figure 2.4 shows reflectance spectra of the three materials as they would appear to the hyperspectral AVIRIS sensor. The gaps in the spectra are wavelength ranges at which the atmosphere absorbs so much light that no reliable signal is received from the surface.

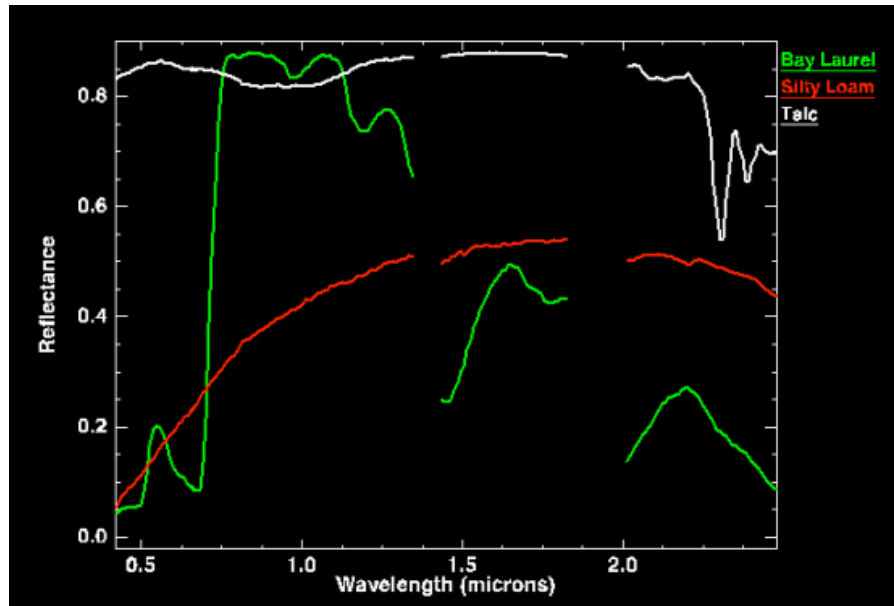


Figure 2.4: Reflectance spectra of the three materials as they would appear to the hyperspectral AVIRIS sensor [2]

Although most hyperspectral sensors measure hundreds of wavelengths, it is not the number of measured wavelengths that defines a sensor as hyperspectral. Rather it is the narrowness and contiguous nature of the measurements. For example, a sensor that measured only 20 bands could be considered hyperspectral if those bands were contiguous and, say, 10 nm wide. If a sensor measured 20 wavelength bands that were, say, 100 nm wide, or that were separated by non-measured wavelength ranges, the sensor would no longer be considered hyperspectral. Standard multispectral image classification techniques were generally developed to classify multispectral images into broad categories. Hyperspectral imagery provides an opportunity for more detailed image analysis. For example, using hyperspectral data, spectrally similar materials can be distinguished [11] and sub-pixel scale information can be extracted. To fulfill this potential, new image processing techniques have been developed.

2.4 Sensor types

A sensor is defined as an instrument capable of measuring electromagnetic radiation. There are hand held sensors (e.g., field spectrometer) which can be used for reference measurements during field work or in the laboratory. Most sensors used in remote sensing are mounted on an air- or spaceborne platform.

The first satellite system to provide near global coverage of the earth's surface on a regular and predictable basis was the Landsat earth resources satellite system. The first three Landsats had identical orbit characteristics, as summarised in Table 2.1. The orbits were near polar and sun synchronous meaning the orbital plane precessed about the earth at the same rate that the sun appears to move across the face of the earth. In this manner data was collected at about the same local time on every pass.

Table 2.1: Landsat 1, 2, 3 orbit characteristics [1]

Orbit:	Sun synchronous,near polar and nominal 9:30 am descending equatorial crossing; in- clined at about 99deg to equator
Altitude:	920km (570 mi)
Period:	103 min
Repeat Cycle:	14 orbits per day over 18 days (251 revolu- tions)

Most past and current hyperspectral sensors have been airborne, with two recent exceptions: NASA's Hyperion sensor on the EO-1 satellite, and the U.S. Air Force Research Lab's FTHSI sensor on the MightySat II satellite. Several new spacebased hyperspectral sensors have been proposed recently. Unlike airborne sensors, space-based sensors are able to provide near global coverage repeated at regular intervals. Therefore, the amount of hyperspectral imagery available should increase significantly in the near future as new satellite-based sensors are successfully launched.

Table 2.2: Current and Recent Hyperspectral Sensors and Data Providers [2]

Sensors	Manufacturer	Number of Bands	Spectral Range
AVIRIS (Airborne Visible Infrared Imaging Spectrometer)	NASA Jet Propulsion Lab makalu.jpl.nasa.gov/	224	0.4 to 2.5 μm
HYDICE (Hyperspectral Digital Imagery Collection Experiment)	Naval Research Lab	210	0.4 to 2.5 μm
FTHSI on MightySat II	Air Force Research Lab www.vs.afrl.af.mil/TechProgs/MightySatII	256	0.35 to 1.05 μm
Hyperion on EO-1	NASA Goddard Space Flight Center eo1.gsfc.nasa.gov	220	0.4 to 2.5 μm
PROBE-1 Earth Search Sciences Inc.	www.earthsearch.com	128	0.4 to 2.5 μm
CASI (Compact Airborne Spectrographic Imager)	ITRES Research Limited www.itres.com up to	228	0.4 to 1.0 μm

2.5 Applications of Hyperspectral Images

Hyperspectral imaging provides researchers the ability to see more than what is visible to the human eye. By accessing the entire electromagnetic spectrum, the sensors are able to image a specific wavelength range, or spectral band, and combine images of multiple bands into a single scene. Hyperspectral imaging systems are used around the world in a variety of laboratory, industrial, and airborne applications. Some of the applications of hyperspectral images are given below:

1. **Remote Sensing:** In remote sensing technology it is very important to distinguish earth surface features, each features have different spectrum band. Multispectral satellite can capture image up few bands for example Landsat 7 have 8 bands. In contrast, hyperspectral imaging satellite can capture earth surface in more than 200 bands which helps scientist to differentiate objects that were not possible in multi spectral imaging because of spectral resolution.
2. **Precision Agriculture:** By using the hyperspectral image and plotting the reflectance spectrum one can conclude that whether those seed are viable or not viable. Seed might be looking same through naked eyes but its viability will be trace down by the hyperspectral image.
3. **Biotechnology:** Hyperspectral imaging is rapidly becoming an essential tool to acquire the data that can be used in the laboratory. They are mostly used in the study of the wound analysis, fluorescence microscopy, and cell biology.
4. **Ground Object Detection:** Hyperspectral images are blessing in the field of ground object detection. Ground object can be detected more accurately with the help of hyperspectral image analysis than all other conventional methods. In this research work, a subspace detection method of hyperspectral data is proposed for ground object identification.
5. **Environmental Monitoring:** Hyperspectral imaging is becoming widely popular for tracking changes in the environment. It is commonly used to understand surface CO₂ emissions, map hydrological formations, tracking pollution levels, and more.
6. **Food Quality:** Hyperspectral images can offer information about the chemical composition of food products. Therefore, the hyperspectral sensor systems are considered as

powerful in-line inspection tools for increasing the quality and safety of food products such as poultry, fruits and vegetables.

7. **Security and Defense:** Hyperspectral imaging technology is ideally suited for defense applications and target identification, allowing identification of man-made materials, distinguishing them from natural surroundings.
8. **Medical Sciences:** For non-invasively diagnostic medical applications, hyperspectral images are used to get spectral information relating to the patient, tissue sample, or disease condition.

Chapter 3

Literature Review

3.1 Introduction

This chapter discusses the recent studies on hyperspectral image analysis and feature reduction approaches. Remotely sensed hyperspectral images provide rich information to meet the requirements of various applications. But the high-dimensionality of the data also introduces many problems including difficulty in processing, computational cost and classification of objects. Moreover, insufficient number of training samples compared to the dimensionality results in a reduction in the classification accuracy. This is known as the curse of dimensionality or the Hughes phenomenon. These issues have fostered the need to develop techniques to effectively reduce the number of features before classification.

Different methods for feature reduction have been developed in the remote sensing literature. This chapter will discuss and review the current feature reduction approaches. This will be followed by the classification techniques available for hyperspectral data in the current literature.

3.2 Dimensionality Reduction

Dimensionality reduction is the process of obtaining a reduced dimension by transforming the data or by removing the least significant image bands from the data. They are known as feature extraction and feature selection respectively. Hyperspectral data generated by sensors contain rich information for a wide range of applications. But a specific application may not find all the measurements to be useful or important. In addition, the original feature space may not be the most effective space for representing the data. In such cases, dimensionality reduction provides

the following benefits:

- It helps in data compressing and thus reducing the storage space required.
- It also fastens the time required for performing different computations. Less dimensions leads to less computing, also less dimensions can allow usage of algorithms unfit for a large number of dimensions.
- It takes care of multi-collinearity that improves the model performance by removing redundant features.
- Reducing the dimensions of data to 2D or 3D may allow us to plot and visualize it precisely. You can then observe patterns more clearly.

Mining significant features from the data includes feature generation, feature selection (FS), and feature extraction (FE) as illustrated in Figure 3.1.

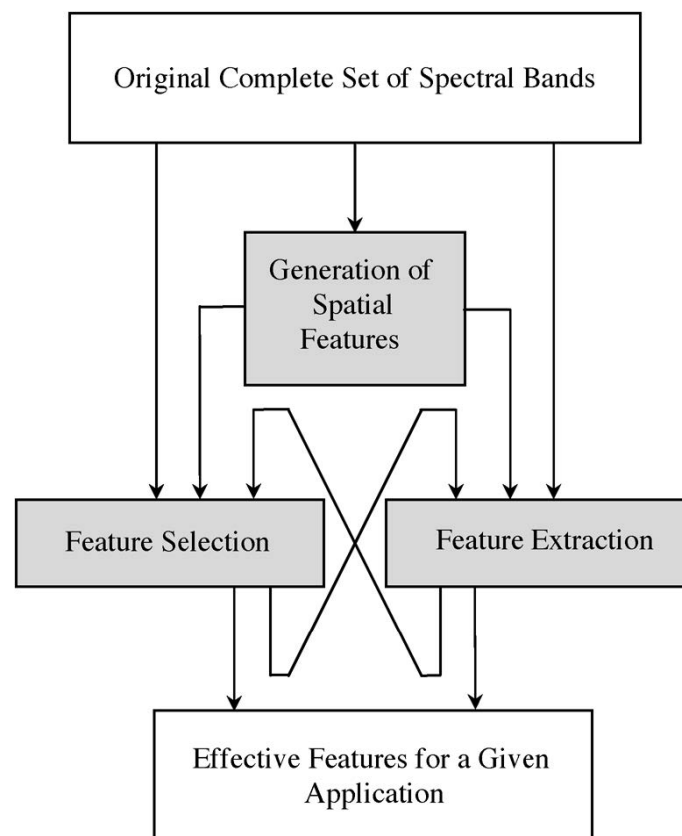


Figure 3.1: Possible operations for feature mining (only one path is used for any specific approach). [12]

A lot of effort has been given in researching this issue since hyperspectral data became available in the late 1980s. The feature mining techniques which have been developed include supervised and unsupervised, parametric and nonparametric, linear and nonlinear methods, which all seek to identify the informative subspace.

3.2.1 Feature Extraction

Feature extraction is a general term that refers to techniques used for generating a feature (sub)set from the vector of available spectral bands. It starts from an initial set of measured data and builds derived features expected to be non-redundant and informative. It facilitates the subsequent learning and generalization steps, and in some cases leads to better human interpretations. In feature extraction, new features are created by transforming the initial set of raw input data into a new feature space. This results in more manageable groups or features for processing, while still accurately and completely describing the original data set.

3.2.2 Feature Selection

feature selection, also known as variable selection, attribute selection or variable subset selection, refers the process of selecting a subset of relevant features based on some criteria for use in model construction [12]. Feature selection techniques are different from feature extraction. Feature extraction produces new features from functions of the original features. On the other hand, feature selection returns a subset of the features.

If the number of samples for a given dataset is much less than the features, then machine learning becomes particularly difficult, since the search space will be sparsely populated. As a result, the model will not be able to differentiate accurately between relevant data and noise. Feature selection enables the machine learning algorithm to train faster and also reduces overfitting. There are two major approaches to feature selection. They are Individual Evaluation and Subset Evaluation. Individual Evaluation is the ranking of the features. In Individual Evaluation, the weight of an individual feature is assigned according to its degree of relevance. In Subset Evaluation, candidate feature subsets are constructed using search strategy.

The general procedure for feature selection has four main steps [13] as shown in Figure 3.2.

- Subset Generation
- Evaluation of Subset

- Stopping Criteria
- Result Validation

Subset generation is a heuristic search in which each state specifies a candidate subset for evaluation in the search space. Two basic issues determine the nature of the subset generation process. First, successor generation decides the search starting point, which influences the search direction. To decide the search starting points at each state, forward, backward, compound, weighting, and random methods may be considered. Second, search organization is responsible for the feature selection process with a specific strategy, such as sequential search, exponential search or random search. A newly generated subset must be evaluated by a certain evaluation criteria. Therefore, many evaluation criteria have been proposed in the literature to determine the goodness of the candidate subset of the features. Based on their dependency on mining algorithms, evaluation criteria can be categorized into groups: independent and dependent criteria. Independent criteria exploit the essential characteristics of the training data without involving any mining algorithms to evaluate the goodness of a feature set or feature. And dependent criteria involve predetermined mining algorithms for feature selection to select features based on the performance of the mining algorithm applied to the selected subset of features. Finally, to terminate the selection process, stopping criteria must be specified. Feature selection process finishes at validation procedure. It is not a part of feature selection process, but feature selection method must be validated by carrying out different tests and comparisons with previously established results or comparison with the results of competing methods using artificial datasets, real world datasets, or both.

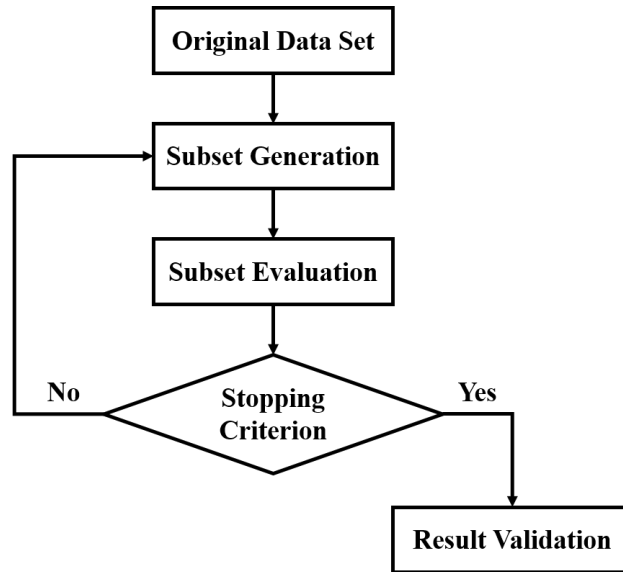


Figure 3.2: Four key steps for the feature selection process

3.2.3 Feature Relevancy and Redundancy

The optimal feature subset is a subset of all relevant features. Therefore, the relevance of the features must be properly defined according to their relevance. In many literatures, a feature's degree of relevance to the target variable is stratified by three qualifiers: irrelevant, weakly relevant, and strongly relevant. Weakly relevant features can be further divided into redundant and non-redundant ones [14]. An optimal classifier uses all strongly relevant features and non-redundant weakly relevant features if the true underlying distribution is known. Fig. 3.3 illustrates the feature relevance hierarchy and an optimal feature subset.

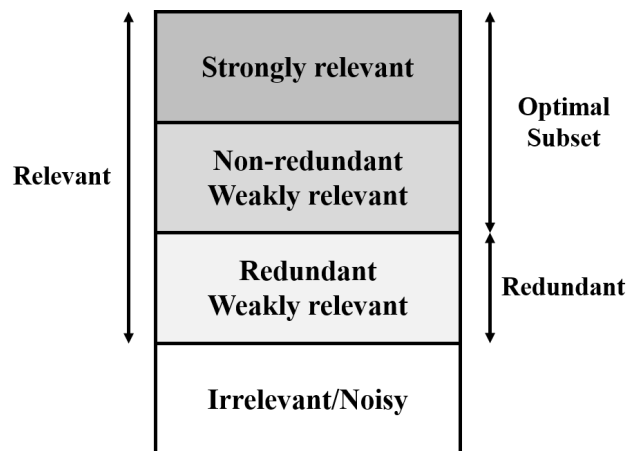


Figure 3.3: Relevance and redundancy of feature

Various definitions have been proposed to answer the question "relevant to what?". In [15], a greedy search approach is applied to choose the most relevant and less redundant features to the class labels.

3.2.4 General Approaches for Feature Selection

There are three general classes of feature selection algorithms [13]: filter methods, wrapper methods and embedded methods.

1. **Filter method:** Filter feature selection methods apply a statistical measure to assign a scoring to each feature. The features are ranked by the score and either selected to be kept or removed from the dataset. The methods are often univariate and consider the feature independently, or with regard to the dependent variable. Some examples of some filter methods include the Chi squared test, information gain and correlation coefficient scores.

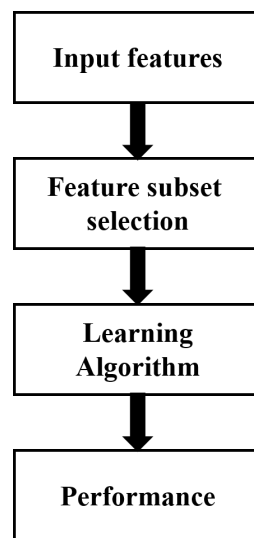


Figure 3.4: Filter Method for feature selection

However, filter methods tend to select redundant variables because they do not consider the relationships between variables. Therefore, they are mainly used as a pre-process method.

2. **Wrapper method:** Wrapper methods consider the selection of a set of features as a search problem, where different combinations are prepared, evaluated and compared to

other combinations. A predictive model is used to evaluate a combination of features and assign a score based on model accuracy.

The search process may be methodical such as a best-first search, it may stochastic such as a random hill-climbing algorithm, or it may use heuristics, like forward and backward passes to add and remove features. An example of a wrapper method is the recursive feature elimination algorithm.

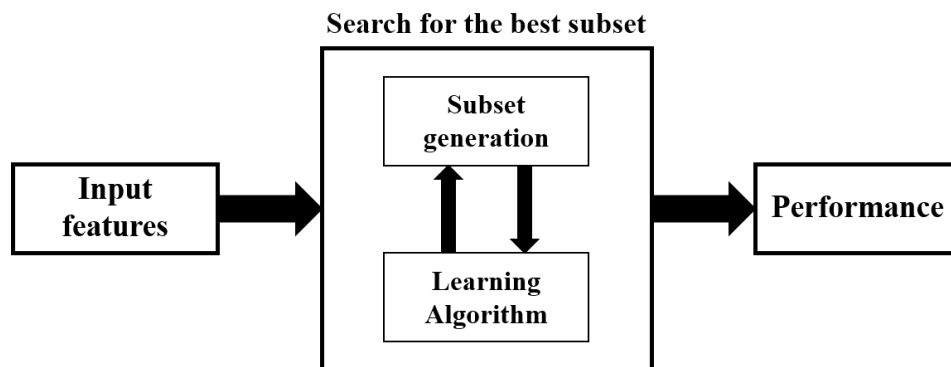


Figure 3.5: Wrapper Method for Feature selection

3. **Embedded method:** Embedded methods combine the qualities of filter and wrapper methods. They learn which features best contribute to the accuracy of the model while the model is being created. They are implemented by algorithms that have their own built-in feature selection methods.

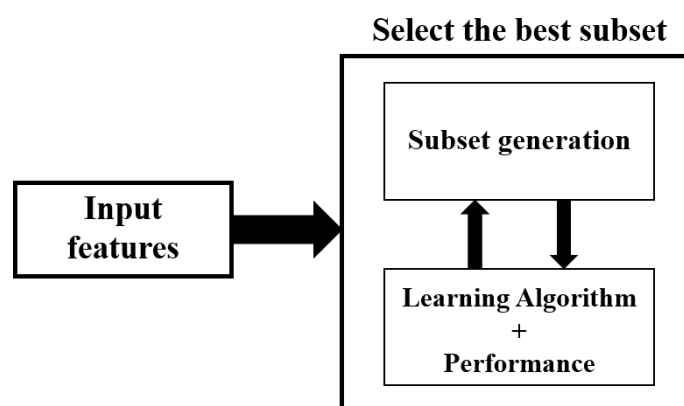


Figure 3.6: Embedded Method for Feature selection

3.3 Related Works

Many feature reduction methods have been proposed in the literature, and their comparative study is a difficult task. Without knowing the relevant features in advance of the real data set, it is very tough to find out the effectiveness of the feature reduction methods, as data sets may include many challenges such as the huge number of irrelevant and redundant features, noisy data, and high dimensionality in term of features or samples. Therefore, the performance of the feature reduction method relies heavily on the performance of the learning method. There are many performance measures mentioned in the literature such as accuracy, computer resources, ratio of feature selection etc. Most researchers have come to the agreement that there is no so-called "best method". Therefore, the new methods for feature reduction are constantly increasing to tackle the specific problem as mentioned above with different strategy.

- Ensuring a better behavior of feature selection using an ensemble method
- Combining with other techniques such as tree ensemble and feature extraction [16]
- Reinterpreting existing algorithms
- Creating a new method to deal with still-unresolved problems
- Combining several feature selection methods

The analysis in [3] suggests that feature selection may be a valuable procedure in preprocessing hyperspectral data for classification by the widely used SVM classifier. In the analysis of hyperspectral data, feature selection is preferred over feature extraction for dimensionality reduction [16], [17]. Feature extraction methods involve transforming the data and hence, crucial information may become compromised or distorted. In contrast, feature selection methods attempt to discover a subset of features which capture the core characteristics of the data, while possessing sufficient capacity to discriminate between classes. Hence, they have the advantage of preserving the relevant original information of the data. There are various studies which establish the usefulness of feature selection in hyperspectral data classification. Various feature selection methods for hyperspectral data have been introduced in current literature such as the Correlation based Feature Selection (CFS), SVM Recursive Feature Elimination (SVM-RFE), Minimum Redundancy Maximum Relevance (MRMR) [18], feature selection and Random Forests [19]. Two main feature extraction techniques in the field of hyperspectral image

analysis are Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA). Several modification for PCA have also been made proposed such as Segmented-PCA (SPCA) and Folded-PCA (FPCA) for better outcome. In [20], a band prioritization scheme based on Principal Component Analysis (PCA) and classification criterion is presented.

Mutual information is a widely used quantity in various feature selection methods. In [21] and [22] uses mutual information to extract most informative features from the original huge data set. In a general setting, features are ranked based on the mutual information between the spectral bands and the reference map(also known as the ground truth). In [6], mutual information is computed using the estimated reference map obtained by using available a priori knowledge about the spectral signature of frequently-encountered materials. In [15], normalized mutual information is computed between class reference and the bands obtained via PCA to select relevant features for better classification accuracy. Cross cumulative residual entropy (CCRE) is another feature selection approach that takes into account the joint and marginal cumulative distribution of sample and class label data. In [23], a combination of PCA and CCRE is proposed for hyperspectral image classification.

Table 3.1: Categories of FS and FE techniques

	Feature Selection	Feature Extraction
Knowledge-based	Depends on spectral characteristics.	Vegetation Index: NDVI; MSAVI2; GA-SVI. Water Index: NDWI.
Supervised (Parametric)	Bhattacharyya distance (B-dis); Jeffries-Matusita (J-M) distance and extended; J-M distance for spatially invariant features; J-M distance for visual inspection and quantitative band selection; Divergence; Divergence on class based PCA data; Cluster space separability; Abundance covariance; Kernel dependence.	Linear Discriminant Analysis (LDA); Regularized LDA; Kernel local Fisher discriminant analysis; Generalized Discriminant Analysis; Canonical Analysis; Combining subsets of adjacent correlated bands into a smaller number of features; Weighted sum of the bands grouped based on B-dis; Averaging contiguous groups of bands based on J-M.
Supervised (Nonparametric)	Mutual Information (MI); Spectral Angle Mapper (SAM); Canonical correlation Measure; Feature weighting.	Decision Boundaries (DBFE); Nonparametric Discrimination Analysis (NDA); Nonparametric Weighted (NWFE); Kernel NWFE; Cosine-based nonparametric (CNFE); double nearest proportion (DNP); Linear Embedding (LE).

Table 3.2: Categories of FS and FE techniques (Cont.)

	Feature Selection	Feature Extraction
Unsupervised	Signal Noise Ratio; Geometric-based representative bands (G-FS); A representative band from each cluster of bands; Dissimilar bands based on linear projection; Minimizing dependency between selected bands using Mutual Information (MI).	PCA; Noise Adjusted PCA or MNF; Kernel PCA (KPCA); Independent Component Analysis (ICA); KICA; Discrete Wavelet Transform; Multiples source feature extraction; Fast ISOMAP; Enhanced ISOMAP; Projection Pursuit (PP); Hurst & Lyapunov Exponents FE; Locally Linear Embedding (LLE); MI based band averaging (Band-Clust); Weighted sum of each cluster of bands, generated using MI and Kullback-Leibler divergence measures; Greedy modular eigenspaces.

3.4 Image Classification

3.4.1 Definition

Image classification refers to analyzing the pixels in a digital image and categorizing them into one of several land cover classes or themes. The categorized data may then be used to generate thematic maps of the land cover present in an image. Generally, hyperspectral data are used to perform the classification where the spectral pattern present within the data for each pixel is used as the numerical basis. The objective of image classification is to identify and portray the features occurring in an image in terms of the object or type of land cover these features actually represent on the ground.

3.4.2 Classification Techniques

Digital image classification techniques group pixels to represent land cover features. Land cover could be forest, urban, agricultural and other types of features. There are two main image classification techniques.

- Unsupervised Classification
- Supervised Classification

Unsupervised Classification: The unsupervised approach attempts to identify spectrally homogeneous clusters of pixels within the image. It results in spectral groupings that may have an unclear meaning from the user's point of view. Having established these, the analyst then tries to manually associate an information class with each group. The unsupervised classification is often referred to as clustering and results in statistics that are for spectral, statistical clusters. There are different unsupervised algorithms available, such as K-means and ISODATA.

Unsupervised classification is a method which examines a large number of unknown pixels and divides into a number of classes without requiring any analyst-specified training data. The basic premise is that values within a given cover type should be close together in the measurement space (i.e. have similar gray levels), whereas data in different classes should be comparatively well separated (i.e. have very different gray levels).

The classes obtained from unsupervised classification are spectrally separable based on natural groupings of the image values and their identity will not be known initially. Therefore the clas-

sified data need to be compared to some form of reference data (such as larger scale imagery, maps, or site visits) to determine the identity and informational values of the spectral classes.

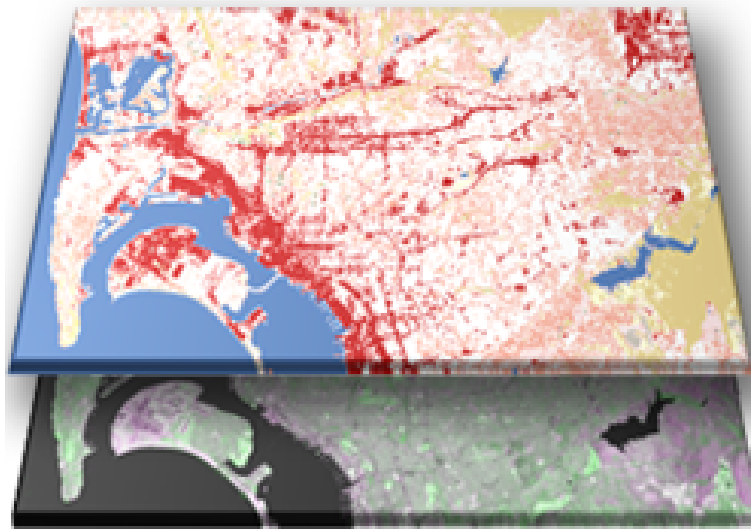


Figure 3.7: Unsupervised Classification [24]

Systems developed in recent years are using clustering procedures that are extremely fast and require little in the nature of operational parameters. For this reason, unsupervised classification is becoming increasingly popular in agencies involved in long term GIS database maintenance. The reason is that there are now systems that use clustering procedures that are extremely fast and require little in the nature of operational parameters. With only a general familiarity GIS analysis can be trained with remote sensing to undertake classifications that meet typical map accuracy standards. Using suitable ground truth accuracy assessment procedures, this tool can provide rapid means of producing quality land cover data on a continuing basis.

Supervised Classification: In the supervised approach to classification the pixel categorization process is supervised by the image analyst. Numerical descriptors of the various land cover types present in the scene are specified to the computer algorithm for classification. Representative sample sites of known cover types, called training areas or training sites, are used to compile a numerical interpretation key that describes the spectral attributes for each feature type of interest. Each pixel in the data set is then compared numerically to each category in the interpretation key and labeled with the name of the category it looks most similar to. The final result is in the form a thematic map where each pixel has a fixed label or class assigned to it.

The common supervised classification algorithms are maximum likelihood [25] and support vector machine [26].

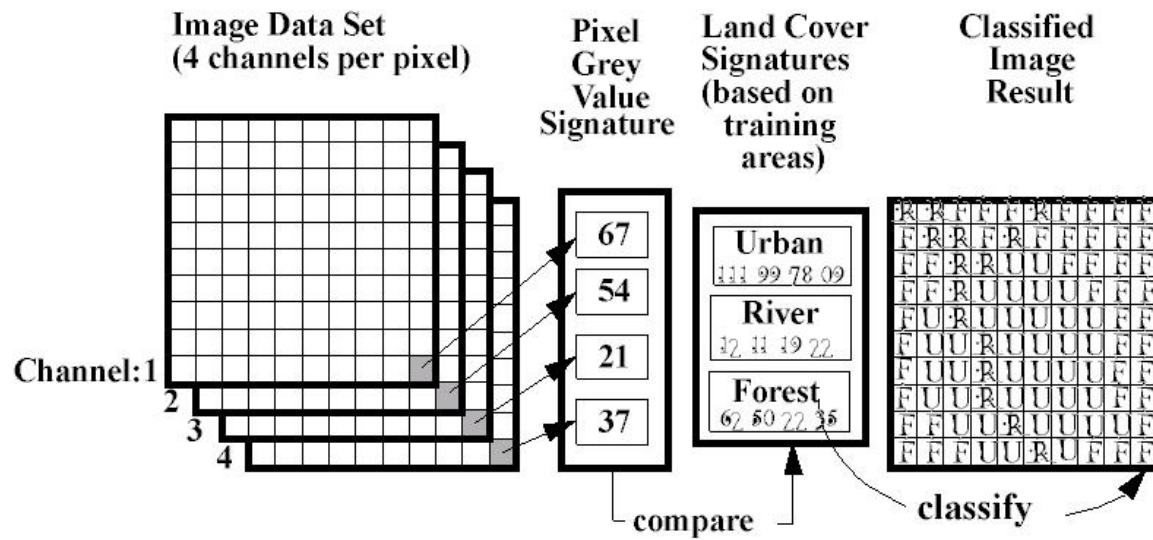


Figure 3.8: Supervised Classification [27]

Chapter 4

Subspace Detection

4.1 Introduction

As discussed in Chapters 1 and 2, hyperspectral sensors capture land cover reflectance in hundreds of narrow contiguous spectral bands and form a large data cube [1]. Thus, they provide better detail on the spectral variations of targets compared to conventional multispectral systems. However, this enormous data cube containing rich information also presents a major challenge in regard to finding the most significant features for classification. This chapter presents an approach which generates new features from the complete set of original spectral bands using Folded Principal Component Analysis (FPCA) and selects an effective subspace based on the required information. Mutual Information (MI) and Cross Cumulative Residual Entropy (CCRE) are used separately to select the relevant features. Since FPCA method has been developed based on the idea of PCA, the chapter begins with a brief overview of PCA.

4.2 Principal Component Analysis (PCA)

Principal component analysis (PCA) is a method used to emphasize variation and bring out strong patterns in a dataset. It is often used to make data easy for exploration and visualization. PCA simplifies the complexity in high-dimensional data while retaining trends and patterns. It does this by transforming the data into fewer dimensions, which act as summaries of features. Consequently, a number of possibly correlated features is transformed into a smaller number of uncorrelated features called principal components (PC).

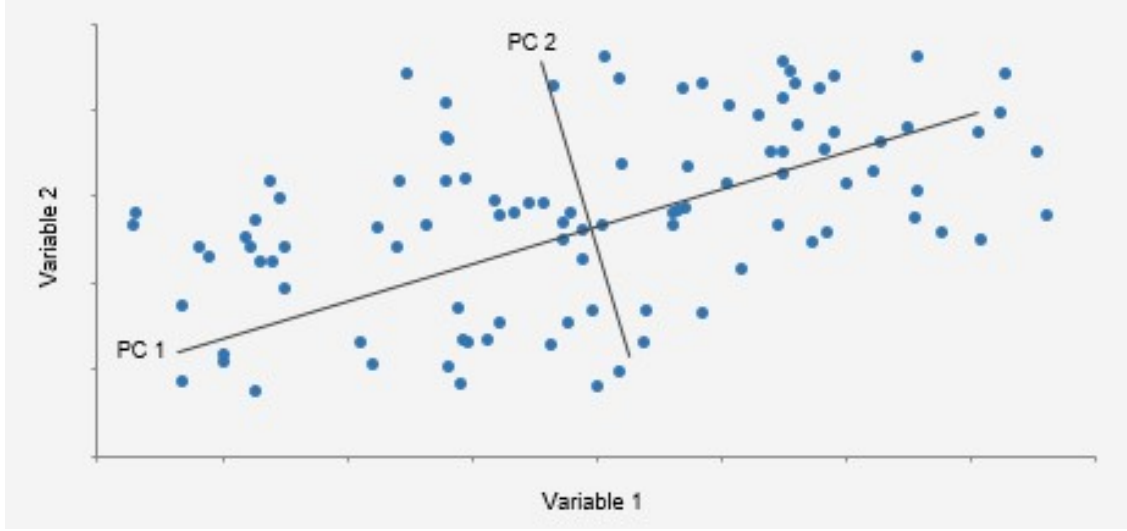


Figure 4.1: Principal component analysis [28]

PCA is useful in hyperspectral image analysis as the neighboring bands of hyperspectral images are highly correlated and often convey similar information about the land cover object. It performs the eigenvalue decomposition of covariance matrix of the input hyperspectral image. The new transformed uncorrelated images are called principal component (PCs). Let $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ be the bands of the input data cube. This data cube is converted into a 2D matrix $\mathbf{X}$ where each column indicates a band. The covariance matrix is formed by calculating the pairwise covariance of the bands. The covariance between any two bands $\mathbf{X}_i$ ($1 \leq i \leq n$) and $\mathbf{X}_j$ ($1 \leq j \leq n$) is calculated as:

$$cov(\mathbf{X}_i, \mathbf{X}_j) = \frac{\sum_{p=1}^M (X_i^p - \mu_i)(X_j^p - \mu_j)}{M} \quad (4.1)$$

Here M is the total number of pixels in each band and μ_i and μ_j are the means of $\mathbf{X}_i$ and $\mathbf{X}_j$ respectively. From the covariance matrix eigenvalues and eigenvectors are determined. The eigenvectors are ordered according to the highest eigenvalue. The first few eigenvectors contain most of the variance of the data set. The new features can be obtained as:

$$\mathbf{Y} = \mathbf{E}^T \mathbf{X} \quad (4.2)$$

Where $\mathbf{Y}$ is the transformed PCA data and $\mathbf{E}^T$ is the transposed matrix of eigenvectors ordered by the higher eigenvalues. The first few components can be used as the extracted features.

4.3 Folded Principal Component Analysis (FPCA)

Despite being widely used, PCA involves some challenges. If the spatial dimension S of hyperspectral image is very large, usually over 100K, obtaining the covariance matrix becomes difficult and it causes softwares such as MATLAB crashed due to memory management problems. PCA also fails to pick-up the distinct contribution of each band as all the bands are treated equally in computing the covariance matrix. These address these issues FPCA is presented in [29], which is also able to give better classification accuracies compared to that given by PCA .

In FPCA, every mean-adjusted spectral vector is converted to a matrix and a partial covariance matrix is determined from it.

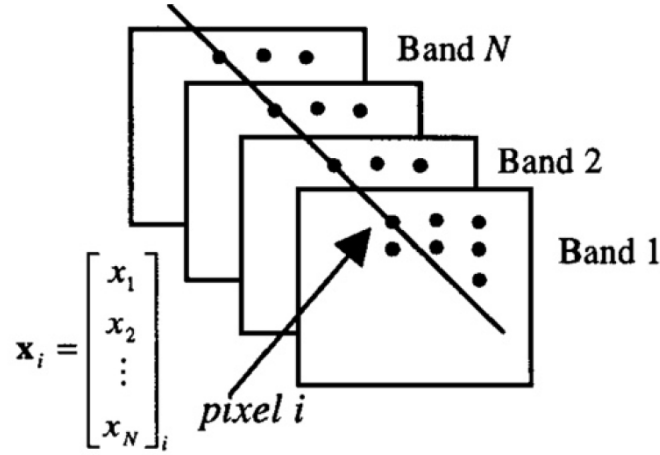


Figure 4.2: Spectral vectors [5]

The covariance matrices are then accumulated for Eigen decomposition and data projection. Thus the whole F bands are folded into H segments (rows), each containing a group of W bands. This is illustrated in Figure 4.3

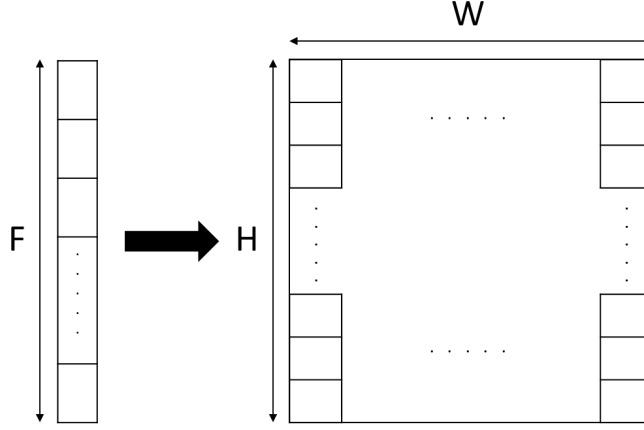


Figure 4.3: Construction of a 2D matrix by folding a 1D spectral vector

Mathematically, Let the spectral vector of a pixel $\mathbf{x}_n$ be defined as $\mathbf{x}_n = [x_{n1} \ x_{n2} \ \dots \ x_{nF}]^T$, where $n \in [1, S]$. The mean-adjusted spectral vectors $\mathbf{I}_n$ are obtained by subtracting the mean spectral vector $\bar{\mathbf{I}}$, i.e.

$$\mathbf{I}_n = \mathbf{x}_n - \bar{\mathbf{I}} \quad (4.3)$$

$$\bar{\mathbf{I}} = \frac{\sum_{n=1}^S \mathbf{x}_n}{S} \quad (4.4)$$

The vector $\mathbf{I}_n = [I_{n1} \ I_{n2} \ \dots \ I_{nF}]^T$ is then converted into matrix $\mathbf{A}_n$ of size $H \times W$, where

$$\mathbf{A}_n = [\mathbf{a}_{n1} \ \mathbf{a}_{n2} \ \dots \ \mathbf{a}_{nH}]_{H \times W}^T \quad (4.5)$$

$$\mathbf{a}_{nh} = [I_{n(1+W(h-1))} \ I_{n(2+W(h-1))} \ \dots \ I_{n(W+W(h-1))}] \quad (4.6)$$

Here, $h \in [1, H]$. The choice of different values of H and W may affect the outcome. For the matrix $\mathbf{A}_n$, the partial covariance matrix is then calculated as follows

$$\mathbf{C}_n = \mathbf{A}_n^T \mathbf{A}_n \quad (4.7)$$

The overall covariance matrix for the entire hypercube is obtained by

$$\mathbf{C}_{FPCA} = \frac{\sum_{n=1}^S \mathbf{C}_n}{S} = \frac{\sum_{n=1}^S \mathbf{A}_n^T \mathbf{A}_n}{S} \quad (4.8)$$

Thus, the size of $\mathbf{C}_{FPCA}$ is $W \times W$. The projection matrix $\mathbf{Z}_n$ corresponding to each folded matrix is calculated after performing Eigen decomposition on this overall covariance matrix as

$$\mathbf{Z}_n = \mathbf{A}_n \mathbf{V} \quad (4.9)$$

Here, $\mathbf{V}$ is the matrix containing eigenvectors sorted based on eigenvalues in descending order. These projection matrices are then used to form an overall projection matrix $\mathbf{Z}$ where each row represents one of the $\mathbf{Z}_n$ matrices. Consequently the columns represent the new features. The first few features are the most significant ones and they can be used to form the reduced feature space $\mathbf{Y}$.

$$\mathbf{Y} = \text{truncated}(\mathbf{Z}) \quad (4.10)$$

4.4 Normalized Mutual Information (nMI)

Mutual information (MI) provides a better and more general criterion for analyzing the relationships between the input variables. It is non-parametric and therefore does not require any assumption about the shape of the distribution of input variables. MI can be defined by Shannon's information theory. If the probability of a random variable x is $P(x)$ then uncertainty of x measured by entropy [30] is:

$$H(x) = -p(x)\log(p(x)) \quad (4.11)$$

Here mutual information can be described as:

$$I(X, Y) = H(X) + H(Y) - H(X, Y) \quad (4.12)$$

Where $H(X)$ and $H(Y)$ are the entropies of X and Y and $H(X, Y)$ is their joint entropy.

MI measures the amount of information one band of hyperspectral image holds about the other band [30]. From eq. 4.12 Mutual information between two image bands $\mathbf{X}$ and $\mathbf{Y}$ can be derived as below [31]:

$$I(\mathbf{X}, \mathbf{Y}) = \sum_{x=1}^L \sum_{y=1}^L p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (4.13)$$

where x, y are the pixel values and L is the maximum pixel values. $P(x), P(y)$ is the marginal and $P(x, y)$ is the joint probability distribution function of $\mathbf{X}$ and $\mathbf{Y}$ respectively [30]. Considering X as the FPCA features and Y as the reference image, the MI can be measured using eq. 4.13. The MI values are normalized between $[0, 1]$ to define normalized Mutual Information (nMI).

4.5 Cross Cumulative Residual Entropy (CCRE)

The cross cumulative residual entropy was introduced in [32] as a method for measuring similarity of information between two input dataset. The CCRE is defined using cumulative residual distributions instead of probability density functions. The reason behind it is that cumulative residual distributions are more robust and regular than probabilistic distributions in presence of noise. The CCRE between two random variables $\mathbf{A}$ and $\mathbf{B}$ is defined as

$$C(\mathbf{A}, \mathbf{B}) = \varepsilon(\mathbf{A}) - E[\varepsilon(\mathbf{A}|\mathbf{B})] \quad (4.14)$$

Here, $\varepsilon(\mathbf{A})$ is the cumulative residual entropy (CRE) of $\mathbf{A}$ and $E[\varepsilon(\mathbf{A}|\mathbf{B})]$ is the expectation of the conditional CRE between $\mathbf{A}$ and $\mathbf{B}$. These terms are given as [23],

$$\varepsilon(\mathbf{A}) = - \int_{R_+} P(|A| > \lambda) \log P(|A| > \lambda) d\lambda \quad (4.15)$$

$$\varepsilon(\mathbf{A}|\mathbf{B}) = - \int_{R_+^N} P(|\mathbf{A}| > a|\mathbf{B}) \log P(|\mathbf{A}| > a|\mathbf{B}) da \quad (4.16)$$

The CCRE of two images $\mathbf{X}$ and $\mathbf{Y}$ can be formulated as [32],

$$C(\mathbf{X}, \mathbf{Y}) = \sum_{u=1}^L \sum_{v=1}^L G(u, v) \log \frac{G(u, v)}{G_X(u)P_Y(v)} \quad (4.17)$$

where L is the maximum pixel value of the images. The joint and marginal cumulative residual distributions are given as [32],

$$G(u, v) = p(X > u, Y = v) \quad (4.18)$$

$$G_X(u) = \sum_{v=1}^L G(u, v) = \sum_{v=1}^L p(X > u, Y = v) \quad (4.19)$$

4.6 Proposed Subspace Detection Method

The proposed method is a combined subspace detection technique which select relevant features for each class of the dataset using folded principal component analysis (FPCA) for feature extraction [4] and cross cumulative residual entropy (CCRE) for features selection [15]. We call this method FPCA-CCRE. First, the unsupervised feature extraction technique FPCA is applied to find uncorrelated dataset in smaller data space. Then cross cumulative residual entropy (CCRE) is applied on the components obtained via FPCA for selecting the relevant

features [33]. Finally support vector machine (SVM) is applied for classification purpose of the data set [7]. Here first SVM uses those selected features of each class to train a model for that specific class and test data to find test result for that specific class. RBF kernel [34] is used in SVM as the relationship between class label C and dataset X_i is nonlinear. K-fold cross validation is applied to find correct parameter C and γ of KSVM for each target class of the dataset. Finally average of those accuracy for each class is the final classification accuracy.

So the steps of the FPCA-CCRE method can be summarized as given below.

1. FPCA is applied first on the original HSI data set. New features are extracted by reducing the dimensionality with FPCA.
2. CCRE is then calculated between the class label data and each FPCA features.
3. Features are arranged in descending order based on their CCRE values.
4. The selected features are assessed for classification.

4.7 Experimental Procedures

4.7.1 Indian pines (92AV3C) Dataset

The performance of the proposed approach was evaluated using the HSI data set captured by AVIRIS sensor over the Indian Pines test site in North-western Indiana [35]. It has 220 bands and 16 classes with spectral resolution of $0.4\text{-}2.5\mu\text{m}$. For this experiment, the bands covering the region of water absorption (104-108, 150-163, 220) are removed as suggested by various researchers [36].

The Indian Pines scene contains two-thirds agriculture, and one-third forest or other natural perennial vegetation. The RGB image along with ground truth is shown in Fig. 4.4 and Fig. 4.5.



Figure 4.4: RGB image of the AVIRIS Indian Pines dataset



Figure 4.5: Ground truth of AVIRIS 92AV3C dataset

4.7.2 FPCA Based Feature Extraction

- **Step 1: Getting input data.**

In order to process hyperspectral image, the 3 dimensional data cube was converted into 2 dimensional matrix where each column represents a band's pixel values. This matrix was then used as input data for FPCA.

$$\text{inputData} = [\text{Image}_{\text{band1}}, \text{Image}_{\text{band2}}, \dots, \text{Image}_{\text{bandF}}]$$

- **Step 2: Mean-adjusting the data.**

The mean value for each dimension was calculated and then subtracted along the corresponding dimension. This produced matrix columns with mean value of 0. This mean subtracted data is represented by adjustedData.

$$\text{AdjustedData} = [\text{Image}_{\text{band1}} - \text{mean}_{\text{band1}}, \text{Image}_{\text{band2}} - \text{mean}_{\text{band2}}, \dots, \text{Image}_{\text{bandN}} - \text{mean}_{\text{bandF}}]$$

- **Step 3: Folding the spectral vectors.**

The rows of AdjustedData represent spectral vectors. Each of these vectors of length F was converted into an $H \times W$ matrix.

- **Step 4: Calculating the partial covariance matrix of the folded matrix.**

In the 4rd step, the partial covariance matrices were calculated from the folded matrices using eq. 4.7

- **Step 5: Calculating the overall covariance matrix.**

The partial covariance matrices were accumulated to form the overall covariance matrix for the whole dataset as given by eq. 4.8 from the folded matrices.

- **Step 6: Calculating the eigenvector and eigenvalues of the covariance matrix.**

From the overall covariance matrix, the eigenvectors and eigenvalues matrix were calculated. This eigenvectors information is very important as they provide information about the local characteristics of input data. Eigenvalues indicate the variance and eigenvectors indicate to which direction.

- **Step 7: Data projection.**

Finally, data projection was performed to obtain the reduced number of features using eq. 4.9 and 4.10 as discussed before.

4.7.3 CCRE Based Feature Selection

- **Step 1: Getting the FPCA data and Class labels**

The FPCA features were taken as well as the ground truth image i.e. the reference image for generating class labels as the inputs for measuring the CCRE values.

- **Step 2: Calculating CCRE between class labels and bands**

To select or rank best features for classification of the dataset CCRE values were measured according to eq. 4.17 between class labels and each band of the dataset.

- **Step 3: Ordering the features based on CCRE values**

The features were then organized in descending order based on the corresponding CCRE values.

4.8 Experimental Results

4.8.1 Feature Extraction Results

In order to extract features from HSI data set, FPCA folds each spectral vector in various ways based on the number of bands in the image. The folding options for applying FPCA on the Indian Pines data set are shown in Table 4.1. The dimension of the original image is reduced to 20 from 200 using FPCA which accumulate approximately 100% cumulative variance. For example, Fig. 3 shows cumulative variance for FPCA-3.

Table 4.1: Folding Options for FPCA

Option	H	W
FPCA-1	2	100
FPCA-2	4	50
FPCA-3	5	40
FPCA-4	8	25
FPCA-5	10	20

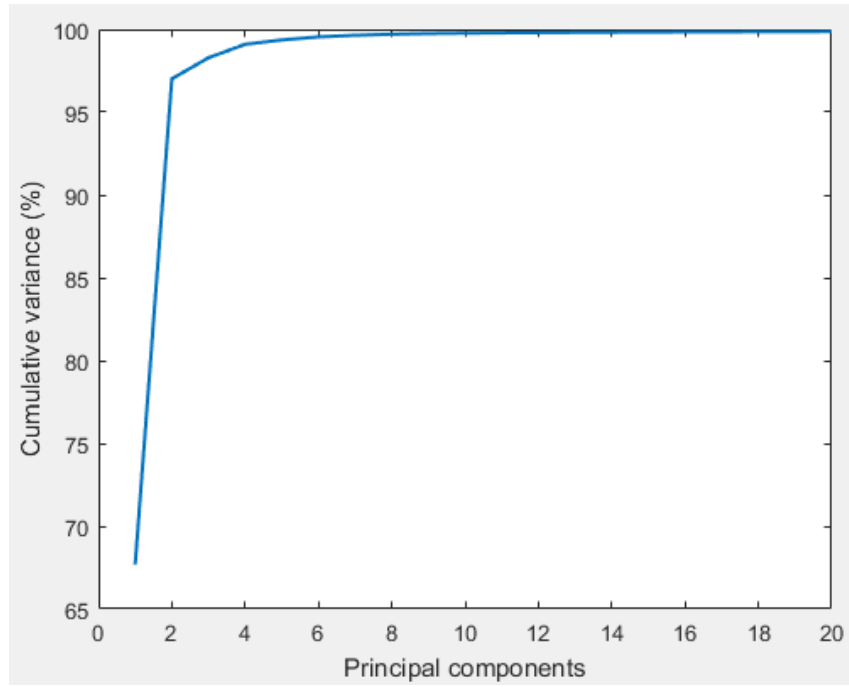


Figure 4.6: Cumulative variance for FPCA-1

FPCA generates principal components with the extraction of the local structure from the hypercube by folding the spectral vectors. Some of the principal components are shown in Figure 4.7

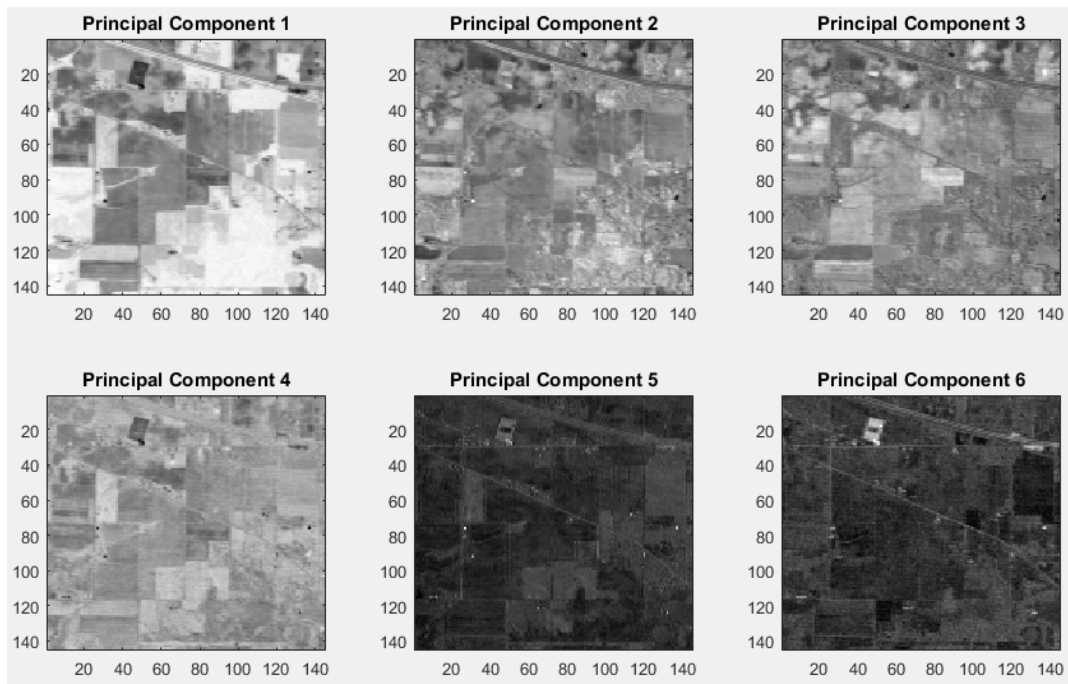


Figure 4.7: Principal components extracted by FPCA-3

Moreover, FPCA requires less memory than PCA for both the data and covariance matrices which can be seen from the comparison in Table 4.2.

Table 4.2: Memory Requirements

Method	Data Matrix Size	Covariance Matrix Size
PCA	$S \times HW$	$HW \times HW$
FPCA	$H \times W$ (Saving factor: S)	$W \times W$ (Saving factor: H^2)

4.8.2 Feature Selection Results

Higher variance does not necessarily guarantees more information. Some features with lower variance may be more informative [37]. To address this issue, CCRE between the class labels and FPCA principal components are calculated as feature selection method. nMI has also been applied separately in the same manner (FPCA-nMI). The sequences of selected features by FPCA-nMI and FPCA-CCRE for all possible folding options are given in Table 4.3

Table 4.3: Selected Features

Method	H $\times$ W Fold	Sequence of Features
FPCA-nMI	2×100	1,3,6,4,5,2,8,12,7,16
	4×50	1,3,4,5,2,6,9,8,7,11
	5×40	2,1,7,5,4,3,11,6,18,14
	8×25	1,3,4,2,6,8,18,16,7,14
	10×20	2,1,3,4,5,14,6,10,8,9
FPCA-CCRE	2×100	1,3,6,4,5,2,8,12,15,7
	4×50	1,4,3,2,8,6,5,9,7,12
	5×40	2,1,7,5,3,11,4,6,10,18
	8×25	1,3,4,2,6,7,8,12,14,18
	10×20	2,1,3,5,4,8,14,6,9,10

It can be seen that instead of variance, principal components are selected by CCRE based on the relevant information. For example, principal component-12 is more informative than principal component-10 extracted by FPCA with the folding option 2×100 (Figure 4.8).

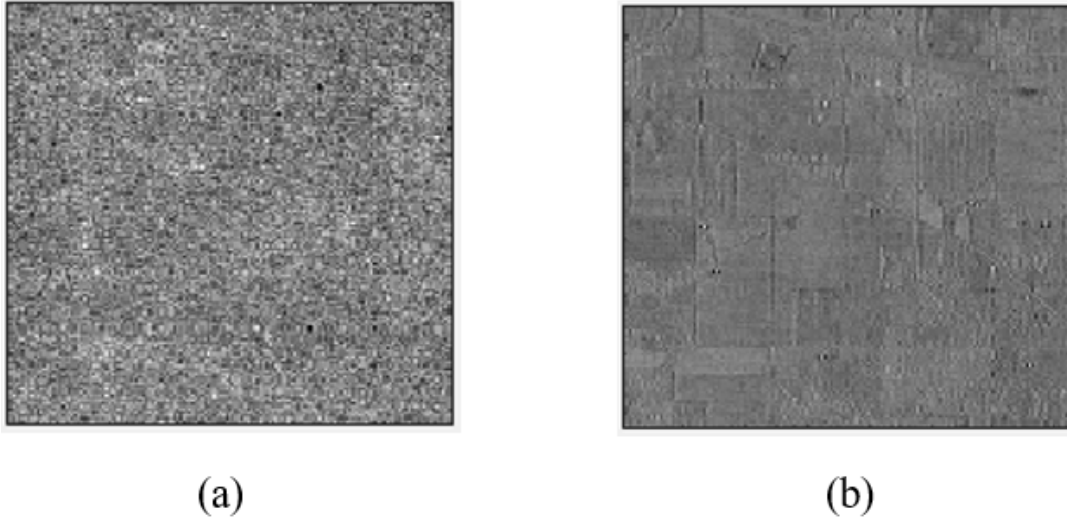


Figure 4.8: (a) Principal component-10 (b) Principal component-12

Two cases of class separation with two features (FPCA 2×100 folding) are shown in Figure 4.9. It can be seen that there are less overlaps among classes when classifying with PC-1 and PC-3 compared to the classification with PC-1 and PC-2.

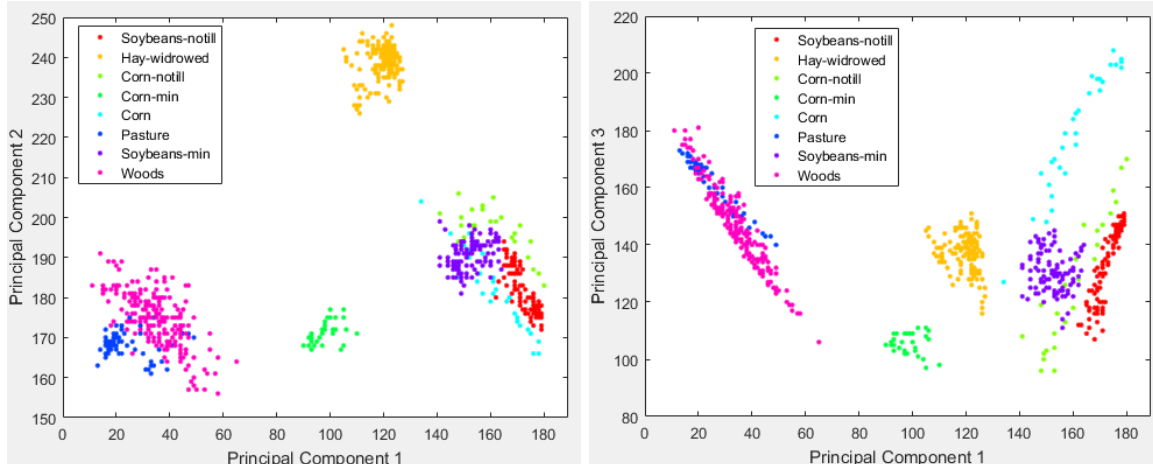


Figure 4.9: Classification with two features

4.8.3 Classification Results

In the experiment, 10 classes are used as reference. The class samples are selected based on the ground truth of the original image [35]. The samples are then divided into train and test samples for the purpose of classification as illustrated in Table 4.4.

Table 4.4: Train and Test Samples

Class Name	Train Samples	Test Samples
Soybeans-notill	104	88
Hay-windrowed	162	136
Soybeans-min	110	135
Woods	208	224
Corn-min	30	24
Grass/Pasture	63	70
Soybeans-clean	22	20
Corn	30	25
Trees	24	15
Corn-notill	25	28

The Selected features are applied to a SVM classifier [11]. The kernel parameters such as best cost parameter C and kernel width γ of RBF kernel function are determined using 10-fold cross validation. SVM is trained with these values of C and γ using the supplied samples. The first 10 selected features for each method are used for classification. The classification results of FPCA, FPCA-nMI and FPCA-CCRE with different folding options are given in Table 4.5.

Table 4.5: Classification Accuracy

Method	H $\times$ W Fold	Classification Accuracy (%)
FPCA	2 $\times$ 100	97.7146
	4 $\times$ 50	96.5922
	5 $\times$ 40	95.5851
	8 $\times$ 25	93.0153
	10 $\times$ 20	93.6915
FPCA-CCRE	2 $\times$ 100	97.9739
	4 $\times$ 50	96.8497
	5 $\times$ 40	95.8693
	8 $\times$ 25	93.5294
	10 $\times$ 20	94.0915
FPCA-nMI	2 $\times$ 100	97.9282
	4 $\times$ 50	96.7938
	5 $\times$ 40	95.7647
	8 $\times$ 25	93.3824
	10 $\times$ 20	94.0654

Fig. 4.10 shows a comparison of classification accuracy among FPCA, FPCA-MI and the proposed FPCA-CCRE approach graphically.

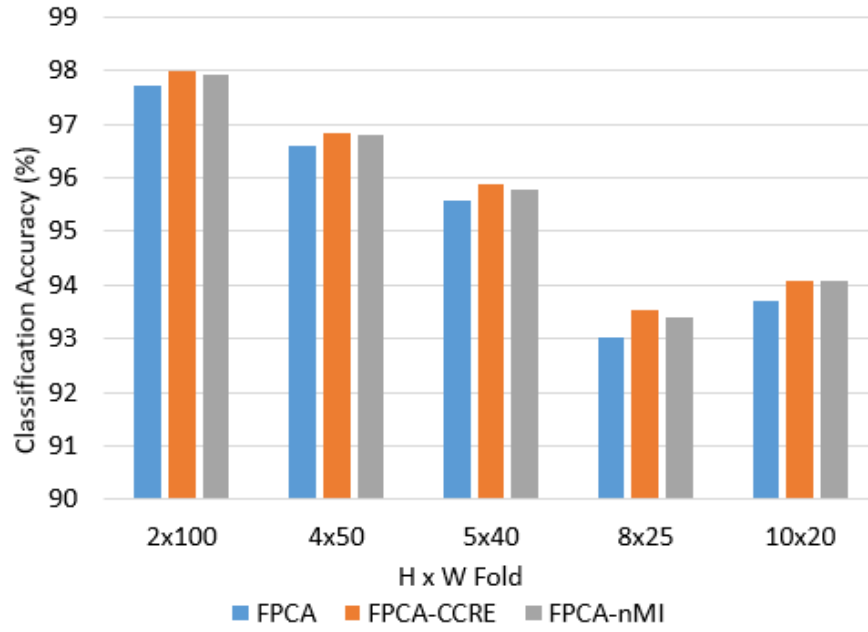


Figure 4.10: Classification results of FPCA, FPCA-nMI and FPCA-CCRE

It can be seen that FPCA-CCRE performs better than FPCA in all the cases. The highest accuracy 97.9739% is achieved by FPCA-CCRE with 2×100 folding which is higher than the accuracy 97.7146% given by FPCA and 97.9282% given by FPCA-nMI. This is because relevant features with local characteristics are selected efficiently by FPCA-CCRE. CCRE tends to outperform nMI since it is comparatively more regular and robust.

Chapter 5

Conclusion and Future Works

5.1 Conclusion

In this thesis work the method is proposed for subspace detection for hyperspectral image classification. The combination of FPCA with CCRE helps to find significant features preserving location characteristics. The performance of the proposed method is evaluated in terms of classification accuracy with SVM. FPCA-CCRE gives better results compared to FPCA and FPCA-nMI. However, if parameters such as folding options, kernel tricks etc. are changed, fluctuation may occur in classification accuracy.

5.2 Future Works

Since the correlations between neighboring bands of hyperspectral image are usually higher than for the bands far apart, the bands can be divided into blocks based on high correlation and FPCA can be applied on each band block separately for better performance.

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